Reg. No.: 22BAI1266

Name :

VIT

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

Continuous Assessment Test II – March 2025

Programme	:	B.Tech. (CSE)	Semester	:	WS2024-25
Course	:	Embedded Systems	Code	:	BCSE305L
			Class Nbr	:	CH2024250501684 CH2024250501686 CH2024250501690 CH2024250501693 CH2024250501705
Faculty	:	Mr. Pugazhenthi R, Dr. Dhanush R, Dr. Sritama Roy, Dr. Manigandan M, Dr. Manimaran P	Slot	:	G1+TG1
Time	:	90 Minutes	Max. Marks	:	50

Answer ALL the questions

Q.No.	Sub. Sec.	Questions	Marks	BT Level
1.		A company is developing a new handheld spot billing machine for use in various industries, such as electricity, water billing, retail, and transportation ticketing. The machine is powered by an ARM Cortex-M3 processor and has multiple connectivity options, including USB, GPRS, and smart card interfaces. However, the company is facing concerns about security vulnerabilities, power efficiency, and real-time performance.	10	L3
	(i)	Analyse the potential security risks associated with the handheld billing machine and propose security measures to mitigate them. (4 Marks)		
	(ii)	How can the device's power consumption be optimized without compromising performance? (3 Marks)		
	(iii)	Compare the ARM Cortex-M3 processor used in this device with AVR, PIC and MCS51 microcontroller families in terms of performance, power efficiency, and suitability for real-time embedded applications. (3 Marks)		
2.		Consider the following C code segment used in an embedded system for processing sensor data	10	L5

```
#include <stdio.h>
#define N 1000
void process_data(int *data, int *result) {
    for (int i = 0; i < N; i++) {
        result[i] = (data[i] * 2) + (data[i] / 3) - (data[i] % 5);
    }
}
```

```
int main() {
  int data[N], result[N];

  // Initialize the data array with some values
  for (int i = 0; i < N; i++) {
    data[i] = i * 2;
  }
  process_data(data, result);

  printf("Processing Complete.\n");
  return 0;
}
```

(i) Apply loop optimization to optimize the given code for an embedded system with limited processing power and memory constraints and provide an optimized version of the code (6 Marks)

(ii) Discuss the improvements in computational efficiency, memory access patterns, and instruction execution cycles in an embedded system environment. (4 marks)

3. Embedded systems, such as autonomous vehicle control and smart traffic management, involve distributed components operating asynchronously. Ensuring fault tolerance and preventing deadlocks is critical for system reliability. 10 L4

(i) Design a model based on Petri Nets to detect deadlocks or livelocks in a distributed real-time embedded system, where multiple nodes (e.g., sensors, controllers, and actuators) must coordinate asynchronously. (5 Marks)

(ii) Analyse the techniques that can be employed within the Petri Net framework to prioritize and schedule transitions in a time-sensitive embedded system to meet hard real-time constraints. (5 Marks)

4. Consider a real-time embedded system that manages an autonomous drone fleet for delivering medical supplies. The drone system needs to schedule tasks such as navigation, obstacle avoidance, battery monitoring, and package handling in real-time. 10 L3

The following periodic tasks are scheduled using the Earliest Deadline First (EDF) algorithm:

Task	Execution Time (C)	Period (P)
T1 (Navigation Update)	2ms	8ms
T2 (Obstacle Avoidance)	3ms	10ms
T3 (Battery Check)	2ms	12ms
T4 (Package Handling)	4ms	20ms

	(i)	Compute the CPU utilization (U) plot (Time Vs. Task) for the given task set. (4 Marks)		
	(ii)	Perform the necessary schedulability test for EDF scheduling and determine whether the system can meet all task deadlines. (3 Marks)		
	(iii)	Suppose if the necessary test fails, suggest possible modifications (such as increasing the period of lower-priority tasks) to make the system schedulable under EDF. (3 Marks)		
5		A company is developing an autonomous vehicle control system that requires real-time processing for various functionalities, including: • Sensor Fusion (T1): Collects and processes data from multiple sensors. • Path Planning (T2): Computes the best possible route while avoiding obstacles. • Vehicle Control (T3): Sends commands to actuators for steering, acceleration, and braking. • Emergency Handling (T4): Detects critical failures and initiates safety protocols.	10	L3
	(i)	Compare and contrast Monolithic, Microkernel, and Hybrid Kernel architectures in the context of autonomous vehicle control. (4 Marks)		
	(ii)	Discuss which architecture is most suitable for handling real-time constraints and critical decision-making. (3 Marks)		
	(iii)	Given that safety is a top priority, explain which kernel would provide the best fault isolation without compromising real-time responsiveness. (3 Marks)		

Course Faculty

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